Initial question: "How do you utilize local storage in a web application?"

Analysis: This question tests a candidate's ability to use the Web Storage API for client-side persistence. Candidates should know how to use localStorage, its methods, and the limitations of local storage. Best practice is to store only small, non-sensitive data (like user preferences) while being mindful of storage limits and synchronous operations. Candidates should also demonstrate awareness of data serialization, browser compatibility, and security by avoiding sensitive information.

Answer: Local storage in a web application is accessed through the `localStorage` object, which is part of the Web Storage API. This API enables developers to store key-value pairs directly in the user's browser, with data persisting even after the browser is closed or the device is restarted. To ensure compatibility, developers should first check if `window.localStorage` exists before attempting to use it.

Data is stored using `localStorage.setItem(key, value)`, retrieved with `localStorage.getItem(key)`, and removed with `localStorage.removeItem(key)`. To clear all data, use `localStorage.clear()`. All stored data must be in string format. If a non-string value is stored directly in localStorage, the value is automatically converted to a string before being stored. For example, storing a number, array, or object results in the following conversions:

- *Number*—`localStorage.setItem("num", 123)` stores "123".
- *Array*—`localStorage.setItem("arr", [1,[2][3])` stores "1,2,3".
- *Object*— `localStorage.setItem("obj", {a: 1})` stores "[object Object]".
- *Boolean, Null, or Undefined*—These are also converted to their string representations, e.g., "true", "null", or "undefined".

This automatic conversion can lead to unexpected results, especially with objects and arrays, because the original structure is lost. To preserve complex data types, objects or arrays should be explicitly serialized using `JSON.stringify()` before storing, and deserialized with `JSON.parse()` when retrieving.